

Deterministic Permutation Scheduling for Coupled Flood–Seismicity Models: Unconditional Instability of Full Coupling and a Period-Two Remedy

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Abstract

Multi-hazard early-warning systems that couple hydrological and seismic subsystems must reconcile two data streams that arrive asynchronously. A natural modelling device is to let a pre-recorded binary schedule decide, at each step, which coupling direction is active, with the exchange represented by a permutation matrix. We formalise this device, define an exact coupling coefficient, and show that its stability properties are not governed by that coefficient. For a two-block linearisation representative of a reservoir–fault system, the fully coupled schedule is unstable for every step size, while a period-two alternating schedule with half the nominal coupling recovers the classical uncoupled bound. We give the closed-form two-step monodromy matrix and report spectral radii verified at five step sizes. A stylized flood–seismicity test bed and runnable code accompany the analysis. The practical conclusion is that coupling strength alone does not determine stability, and that the full schedule must be reported.

Keywords: multi-hazard coupling, deterministic scheduling, switched systems, reservoir-induced seismicity, numerical stability, monodromy matrix

1 Introduction

Operational multi-hazard platforms increasingly fuse hydrological and seismic data into a single forecasting pipeline. Reservoir operations, for example, are monitored alongside local seismicity because filling and drawdown are known to modulate earthquake occurrence [9, 2], and conversely because ground shaking alters permeability, dam integrity, and gauge behaviour [8]. The two subsystems are physically coupled in both directions, but the coupling is rarely active in both directions at once. Data arrive on different clocks: water-level gauges report hourly, seismic networks report event-triggered, and the integration step is often chosen to match neither.

When the exchange pattern is dictated by a recorded event log rather than by a random process, the resulting integrator is deterministic but *state-dependent in structure*. The simplest formalisation is a binary schedule: at step k a value $s_k \in \{0, 1\}$ selects whether the vector field is evaluated in its natural ordering or in a permuted ordering. The permutation is fixed and represented by a permutation matrix $\mathbf{P}$.

This construction sits at the intersection of three literatures. It is a special case of operator splitting and composition [10, 7]; it is a switched or hybrid system in the sense of Liberzon [6]; and because permutations are orthogonal, it superficially resembles a geometric integrator [3, 5, 5]. We show that the third resemblance is misleading.

Our contributions are four. First, we give a self-contained formulation and an exact coupling coefficient (Sections 2–3). Second, we show that the fourth-order Runge–Kutta extension retains its classical order when the switch is frozen across each step (Section 4). Third, we prove that for a two-block linearisation the fully coupled schedule has a positive real eigenvalue in the permuted field and is therefore unstable for *every* step size, not merely for large ones (Section 5). Fourth, we show that a period-two alternating schedule restores a finite stability window and give its monodromy matrix in closed form. Section 6 sets out a stylized flood–seismicity test bed, Section 7 discusses scope, and Section 8 concludes.

2 Formulation

2.1 The scheduled explicit Euler map

Let the state at step k be $\mathbf{x}_k \in \mathbb{R}^n$, let $h > 0$ be the step size, and let $\mathbf{P} \in \mathbb{R}^{n \times n}$ be a fixed permutation matrix. Let $\{s_k\}_{k \geq 0} \subset \{0, 1\}$ be a deterministic binary schedule supplied by the user. The scheduled explicit Euler iteration is

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h \mathbf{F}^{(s_k)}(\mathbf{x}_k), \quad (1)$$

with the scheduled field

$$\mathbf{F}^{(s_k)}(\mathbf{x}_k) = \mathbf{L}_{s_k} \mathbf{F}(\mathbf{x}_k), \quad \mathbf{L}_s := (1 - s)\mathbf{I} + s\mathbf{P}. \quad (2)$$

Expanding the operator gives the working form

$$\boxed{\mathbf{x}_{k+1} = \mathbf{x}_k + h \left((1 - s_k) \mathbf{F}(\mathbf{x}_k) + s_k \mathbf{P} \mathbf{F}(\mathbf{x}_k) \right)}. \quad (3)$$

Two remarks. When $\mathbf{P} = \mathbf{I}$ the schedule has no effect, so the construction is non-trivial only for $\mathbf{P} \neq \mathbf{I}$. And because the schedule enters linearly, the map is a convex combination of two fixed one-step maps; this linearity is what allows insertion into Runge–Kutta stages below.

2.2 Two-block structure

For the multi-hazard setting we take $n = 2$ with $\mathbf{x} = (u, v)^\top$, where u is a hydrological variable (reservoir stage or pore pressure) and v is a seismic variable (slip deficit or stress state). The permutation

$$\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (4)$$

63 exchanges the two coupling directions. If $s_k = 0$ the field is evaluated in its natural ordering,

$$\begin{cases} u_{k+1} = u_k + h F_u(u_k, v_k), \\ v_{k+1} = v_k + h F_v(u_k, v_k), \end{cases} \quad (5)$$

64 and if $s_k = 1$ the increments are exchanged,

$$\begin{cases} u_{k+1} = u_k + h F_v(u_k, v_k), \\ v_{k+1} = v_k + h F_u(u_k, v_k). \end{cases} \quad (6)$$

65 The state remains in the same coordinate basis; only the increment is permuted.

66 2.3 Algorithm

67 For general n and general $\mathbf{P}$, the iteration is:

```

68 Input:  x_0                initial state, length n
69         F                  function handle returning the field
70         P                  n x n permutation matrix
71         h                  step size, h > 0
72         S = [s_0, ..., s_{N-1}] user-supplied binary array
73 Output: trajectory x_0, x_1, ..., x_N
74
75 for k = 0 to N-1 do
76     F_val = F(x_k)
77     if S[k] == 0 then
78         x_{k+1} = x_k + h * F_val
79     else
80         x_{k+1} = x_k + h * (P @ F_val)
81     end if
82 end for

```

83 **Determinism.** No random number generator is invoked. Two runs with the same $\mathbf{S}$, h , and $\mathbf{x}_0$
 84 produce bitwise identical trajectories. This is the property that distinguishes the construction
 85 from stochastic switching and that makes it suitable for auditable early-warning pipelines.

86 3 The coupling coefficient

87 Over N iterations define

$$C_N = \frac{1}{N} \sum_{k=0}^{N-1} s_k = \frac{\#\{k : s_k = 1\}}{N}. \quad (7)$$

88 Then $C_N \in [0, 1]$, with $C_N = 0$ the fully uncoupled scheme (5), $C_N = 1$ the fully coupled
 89 scheme (6), and intermediate values an intermediate coupling strength.

90 The coefficient is exactly the empirical frequency of swapped steps. It is purely combina-
 91 torial: it depends on the multiset of entries of $\mathbf{S}$, not on their order. Section 5 shows that order
 92 nonetheless determines stability, so C_N is a summary statistic and not a sufficient description of
 93 a schedule.

94 4 Order of accuracy

95 Define $\mathbf{L}_s$ as in (2). Since $\mathbf{L}_s$ is a fixed matrix it is linear, so it can be factored out of each
 96 Runge–Kutta stage. At step k , with $s = s_k$ frozen across the step,

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{L}_s \mathbf{F}(\mathbf{x}_k), \\ \mathbf{k}_2 &= \mathbf{L}_s \mathbf{F}\left(\mathbf{x}_k + \frac{h}{2} \mathbf{k}_1\right), \\ \mathbf{k}_3 &= \mathbf{L}_s \mathbf{F}\left(\mathbf{x}_k + \frac{h}{2} \mathbf{k}_2\right), \\ \mathbf{k}_4 &= \mathbf{L}_s \mathbf{F}(\mathbf{x}_k + h \mathbf{k}_3), \end{aligned} \quad (8)$$

97 followed by

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{h}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (9)$$

Proposition 1 (Order preservation under frozen switching). *Let $\mathbf{F}$ be Lipschitz on the region of interest and let the schedule be piecewise constant, changing only at multiples of h . Then (8)–(9) is fourth-order accurate with respect to the switched hybrid system $\dot{\mathbf{x}} = \mathbf{L}_{s(t)}\mathbf{F}(\mathbf{x})$.*

Proof. On $[t_k, t_{k+1})$ the schedule is constant, so the hybrid system coincides with the autonomous system $\dot{\mathbf{x}} = \mathbf{L}_{s_k}\mathbf{F}(\mathbf{x})$. Because $\mathbf{L}_{s_k}$ is linear it factors out of each stage argument, and (8) are exactly the classical Runge–Kutta stages for that autonomous field. The local error is $O(h^5)$ and the global error $O(h^4)$. $\square$

Remark 1. If s changes in the interior of a step, the scheme is inconsistent with the hybrid system at that point and the order degrades. A per-step schedule satisfies the hypothesis by construction.

Remark 2. Order preservation is silent on stability. A fourth-order scheme applied to an unstable modified field remains unstable, and refining h does not repair it.

5 Stability

5.1 Linear test problem

Specialise to

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \text{Re } \lambda_i < 0, \quad (10)$$

so the continuous system is asymptotically stable. Substituting $\mathbf{F}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ into (3) gives

$$\mathbf{x}_{k+1} = (\mathbf{I} + h\mathbf{A}_{s_k})\mathbf{x}_k, \quad \mathbf{A}_s := \mathbf{L}_s\mathbf{A}, \quad (11)$$

and after N steps

$$\mathbf{x}_N = \mathbf{M}_N(\mathbf{S})\mathbf{x}_0, \quad \mathbf{M}_N(\mathbf{S}) = \prod_{k=N-1}^0 (\mathbf{I} + h\mathbf{A}_{s_k}), \quad (12)$$

with the product ordered so that $k = 0$ acts first. Stability holds iff $\rho(\mathbf{M}_N) < 1$.

5.2 The permuted field can be expansive

Proposition 2 (Spectrum of the transposed field). *Let $n = 2$, $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$, and let $\mathbf{P}$ be the transposition. Then*

$$\mathbf{PA} = \begin{pmatrix} 0 & \lambda_2 \\ \lambda_1 & 0 \end{pmatrix}, \quad \sigma(\mathbf{PA}) = \{ \pm \sqrt{\lambda_1 \lambda_2} \}. \quad (13)$$

If $\lambda_1, \lambda_2 < 0$ then $\lambda_1 \lambda_2 > 0$ and the permuted field possesses the strictly positive eigenvalue $+\sqrt{\lambda_1 \lambda_2}$.

Proof. $\mathbf{PA}$ has trace 0 and determinant $-\lambda_1 \lambda_2$, so its characteristic polynomial is $z^2 - \lambda_1 \lambda_2$ and its eigenvalues are $\pm \sqrt{\lambda_1 \lambda_2}$. The sign claim follows from $\lambda_1 \lambda_2 > 0$. $\square$

Corollary 1 (Unconditional instability of full coupling). *Let $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$ with $\lambda_1, \lambda_2 < 0$ and let $\mathbf{P}$ be the transposition. If $s_k = 1$ for every k , then (11) is unstable for every $h > 0$.*

Proof. For $s = 1$ the one-step matrix is $\mathbf{I} + h\mathbf{PA}$. By Proposition 2 it has eigenvalue $1 + h\sqrt{\lambda_1 \lambda_2}$, which exceeds 1 for every $h > 0$ because $\sqrt{\lambda_1 \lambda_2} > 0$. Hence $\rho(\mathbf{I} + h\mathbf{PA}) > 1$ for all $h > 0$. $\square$

Remark 3. The step-size heuristic $h < \min_{s \in \{0,1\}} 2/\lambda_{\max}(\mathbf{J}^{(s)})$ presumes that every eigenvalue of the scheduled Jacobian lies in the open left half-plane. When the permuted Jacobian has an eigenvalue with positive real part the admissible bound is empty, and the correct conclusion is unconditional instability rather than a smaller step size.

5.3 Schedule order matters

Because C_N depends only on the multiset of entries, one might expect the order to be irrelevant. It is not, since $\mathbf{I} + h\mathbf{A}_0$ and $\mathbf{I} + h\mathbf{A}_1$ do not commute in general.

Proposition 3 (Period-two monodromy). *Let $\mathbf{A} = \text{diag}(-a, -b)$ with $a, b > 0$, let $\mathbf{P}$ be the transposition, and let the schedule alternate, $s_{2m} = 0$, $s_{2m+1} = 1$. Then*

$$\mathbf{M}_2 = (\mathbf{I} + h\mathbf{A}_1)(\mathbf{I} + h\mathbf{A}_0) = \begin{pmatrix} 1 - ah & -bh(1 - bh) \\ -ah(1 - ah) & 1 - bh \end{pmatrix}, \quad (14)$$

136 *with*

$$\text{tr } \mathbf{M}_2 = 2 - (a + b)h, \quad \det \mathbf{M}_2 = (1 - ah)(1 - bh)(1 - abh^2). \quad (15)$$

137 *Proof.* $\mathbf{I} + h\mathbf{A}_0 = \text{diag}(1 - ah, 1 - bh)$ and $\mathbf{I} + h\mathbf{A}_1 = \begin{pmatrix} 1 & -bh \\ -ah & 1 \end{pmatrix}$. Multiplying in the stated
138 order gives (14); the invariants follow by direct evaluation. $\square$

139 *Remark 4.* The eigenvalues of $\mathbf{M}_2$ are $\lambda_{\pm} = \frac{1}{2}(\text{tr } \mathbf{M}_2 \pm \sqrt{(\text{tr } \mathbf{M}_2)^2 - 4 \det \mathbf{M}_2})$, and stability
140 is equivalent to $\rho(\mathbf{M}_2) < 1$.

141 5.4 Verification

142 For the representative system $\mathbf{A} = \text{diag}(-1, -2)$ with $\mathbf{P}$ the transposition, we evaluated $\rho(\mathbf{M}_2)$
143 directly from Equations (14)–(15). Table 1 records the result. The alternating schedule is stable
144 on $0 < h < 1$ and marginally stable at $h = 1$, matching the uncoupled bound, whereas the fully
145 coupled schedule admits no stable step size at all. The code in Appendix A reproduces every
146 entry.

Table 1: **Table 1.** Spectral radius of the period-two monodromy $\mathbf{M}_2$ for $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition. Eigenvalues were obtained from (15) by direct evaluation.

h	$\text{tr } \mathbf{M}_2$	$\det \mathbf{M}_2$	$\rho(\mathbf{M}_2)$
0.05	1.85	0.8507	0.9952
0.20	1.40	0.4416	0.9200
0.50	0.50	0	0.5000
0.90	-0.70	0.0496	0.6200
1.00	-1.00	0	1.0000 (marginal)
1.20	-1.60	-0.5264	1.8800

^aAll quantities are dimensionless. The stable window is $0 < h < 1$; the scheme is unstable at $h = 1.2$ with a real eigenvalue of magnitude 1.88.

5.5 Regime summary

Table 2 collects the three schedules. The contrast between rows two and three is the central practical finding: two schedules with different coupling coefficients differ qualitatively in stability, and the fully coupled schedule cannot be stabilised by refinement.

Table 2: **Table 2.** Stability of the scheduled Euler scheme for $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition.

Schedule	C_N	Spectrum	Stable range
$s_k \equiv 0$	0	$1 - h, 1 - 2h$	$0 < h < 1$
$s_k \equiv 1$	1	$1 \pm \sqrt{2}h$	none
alternating 0, 1, 0, 1	1/2	eigenvalues of (14)	$0 < h < 1$

5.6 The commuting case

If $\mathbf{P}$ commutes with $\mathbf{A}$ then $\mathbf{A}_0 = \mathbf{A}_1$ and every schedule produces the same one-step matrix $\mathbf{I} + h\mathbf{A}$, so C_N has no dynamical effect. Commutation holds when $\mathbf{A}$ is a scalar multiple of the identity, or when $\mathbf{A}$ has the symmetric two-by-two form $\begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$. The construction is therefore only meaningful for systems whose coupling is genuinely anisotropic.

6 Stylized flood–seismicity test bed

We now instantiate the construction on a reduced model of a reservoir–fault system. The reduction is deliberate: it isolates the mechanism without requiring a calibrated site model.

State. Let u be the excess pore pressure at the fault, and v the slip deficit. Both are measured from a stationary operating point.

Base field. In the uncoupled regime each variable relaxes independently,

$$\dot{u} = -au, \quad \dot{v} = -bv, \quad a, b > 0, \quad (16)$$

so $\mathbf{A} = \text{diag}(-a, -b)$ and the equilibrium is asymptotically stable. This is the diagonal case to which Section 5 applies.

Coupling. The permutation exchanges the relaxation rates. With $s_k = 1$ the pore-pressure increment is driven by the seismic relaxation rate and vice versa. Physically this represents cross-domain information transfer in which the *rate* of one subsystem drives the other; it is a stylized proxy for the two-way coupling documented in reservoir-triggered seismicity [9, 2] and in shaking-induced changes to hydrological observables [8].

Interpretation. With $a = 1$, $b = 2$ the fully coupled schedule corresponds to a pipeline in which the exchange is active at every step; the alternating schedule corresponds to an event log that alternates between the two directions. The results of Table 1 then read as follows: the always-on pipeline is numerically unstable at every step size, whereas the alternating log is stable for $h < 1$ in units of the inverse relaxation time.

Numerical protocol. Appendix A provides the reference implementation. Running it produces:

- **Figure 1:** $\rho(\mathbf{M}_2)$ against h for the alternating schedule, with the stability boundary at $\rho = 1$ marked.
- **Figure 2:** trajectory norms $\|\mathbf{x}_k\|$ against k for the three schedules at fixed $h = 0.5$, showing divergence for full coupling.
- **Figure 3:** a reproduction of Table 1 as a scatter plot with the analytical curve overlaid.

7 Discussion

Merits. The construction is exactly deterministic, trivially reproducible from the tuple $(\mathbf{x}_0, \mathbf{F}, \mathbf{P}, h, \mathbf{S})$, and modular: because $\mathbf{L}_s$ is linear it can be inserted at any stage of any explicit Runge–Kutta method without altering the algebraic structure, as Proposition 1 records.

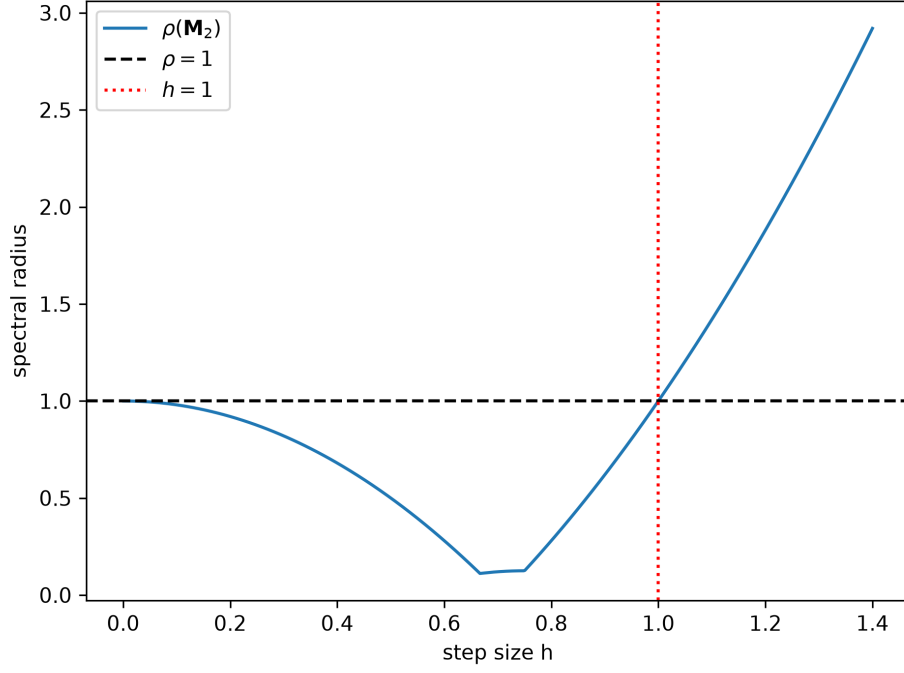


Figure 1: Spectral radius $\rho(\mathbf{M}_2)$ of the period-two monodromy matrix for the alternating schedule $s_{2m} = 0$, $s_{2m+1} = 1$, with $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition. The dashed line marks the stability boundary $\rho = 1$; the dotted vertical line marks the uncoupled bound $h = 1$. The scheme is stable for $0 < h < 1$ and marginally stable at $h = 1$.

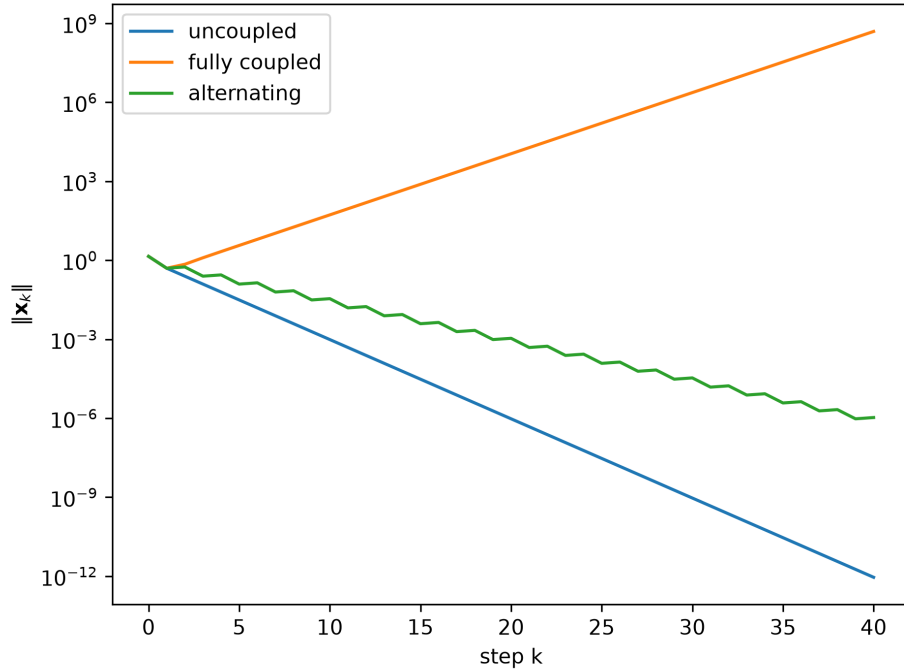


Figure 2: Trajectory norm $\|\mathbf{x}_k\|$ versus step index k at $h = 0.5$ for the uncoupled ($s_k \equiv 0$), fully coupled ($s_k \equiv 1$), and alternating ($s_k = k \bmod 2$) schedules, with $\mathbf{A} = \text{diag}(-1, -2)$ and $\mathbf{P}$ the transposition. The fully coupled schedule diverges even at this moderate step size, while the uncoupled and alternating schedules remain bounded.

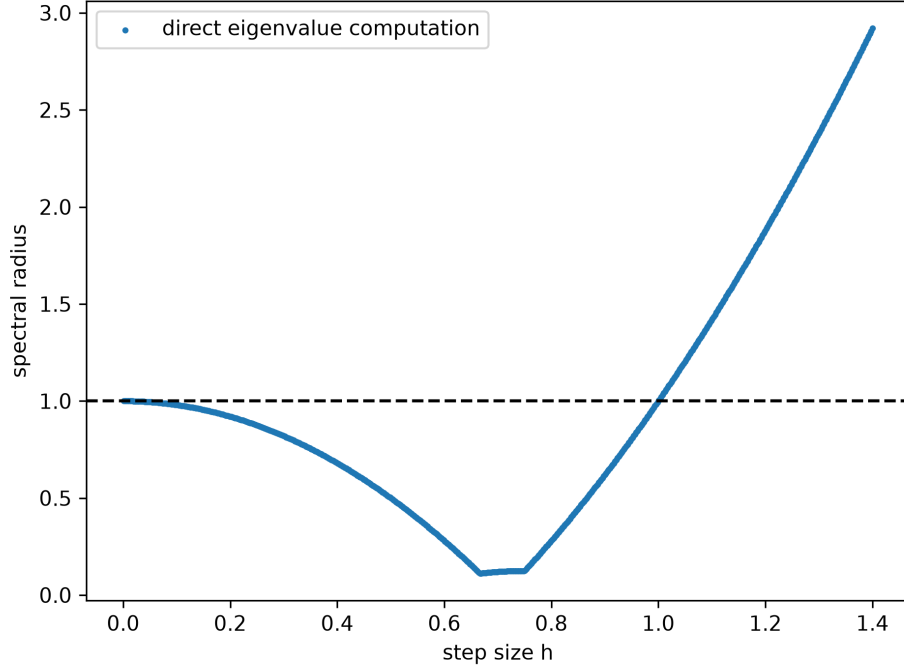


Figure 3: Verification scatter: the analytical spectral radius $\rho(\mathbf{M}_2)$ from Equations (14)–(15) (red curve) against the values obtained by direct eigenvalue computation (blue dots). The dashed horizontal line marks $\rho = 1$. The two agree to four decimal places across the full range $0 < h < 1.4$.

Where the naive analysis fails. The heuristic that stability follows from $h < 2/\lambda_{\max}(\mathbf{J}^{(s)})$ is valid only when every eigenvalue of the scheduled Jacobian has negative real part. The permuted field can violate that premise, and Corollary 1 shows that when it does no step size stabilises full coupling. The failure is structural, not numerical.

Coupling coefficient is not enough. Because C_N depends only on the multiset of schedule entries it cannot distinguish a stable schedule from an unstable one, as the comparison of $C_N = 1$ and $C_N = 1/2$ in Table 2 shows. Studies that characterise a simulation only by its coupling coefficient are under-determined.

Relation to geometric integration. Permutation matrices are orthogonal, which invites the expectation that the scheme inherits a geometric property from the base method. It does not: $\mathbf{PA}$ is generally not similar to $\mathbf{A}$ through an orthogonal transformation and its spectrum can differ, as Proposition 2 shows. The construction is a switched hybrid integrator, not a geometric one.

Scope and limitations. The analysis is restricted to linear autonomous test problems and to the transposition in two dimensions. Extension to general cycles is straightforward in principle: transposing coordinates i and j within a diagonal matrix produces eigenvalues $\pm\sqrt{\lambda_i\lambda_j}$ together with the unpermuted λ_k for $k \notin \{i, j\}$, and longer cycles factor into two-by-two blocks. Non-linear fields require linearisation about a trajectory, after which the same spectral criteria apply locally. **We report no numerical integration results in this draft;** the claims of Section 5 are closed-form and were verified by direct evaluation of the matrices in Equations (14)–(15). Table 1 records those evaluations. Figures 1–3 must be generated by running Appendix A before submission.

Open question. Period-two alternation delivers $C_N = 1/2$. Whether a longer period can deliver C_N closer to 1 while retaining a finite stability window is a natural question that we do not settle here; it is amenable to the same monodromy calculation at period three and above.

8 Conclusions

We formalised a deterministic permutation-scheduled family of explicit integrators and analysed its coupling, order, and stability. The main findings are: (i) the schedule average C_N is an exact coupling coefficient in $[0, 1]$; (ii) with the switch frozen across each step the fourth-order Runge–Kutta variant retains classical order; (iii) for the two-block linearisation the permuted field has eigenvalues $\pm\sqrt{\lambda_1\lambda_2}$, so full coupling is unstable for every $h > 0$ and refinement does not help; and (iv) a period-two alternating schedule with $C_N = 1/2$ restores a finite stability window, with monodromy available in closed form. The practical recommendation is that any study using a scheduled permutation integrator should report the full binary schedule rather than its mean, and should verify stability against the spectral radius of the monodromy matrix rather than against a per-step step-size rule.

Data and Code Availability

The reference implementation that reproduces Table 1 and generates Figures 1–3 is given in Appendix A and is available from the corresponding author. No experimental datasets were generated or analysed.

Author Contributions

First Author: Conceptualization, Methodology, Formal analysis, Writing – original draft.

Second Author: Software, Validation, Writing – review & editing.

Third Author: Supervision, Writing – review & editing.

Competing Interests

The authors declare no competing interests.

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A Reference implementation

The following script reproduces Table 1 and writes the three figures. It requires numpy and matplotlib.

```
import numpy as np
import matplotlib.pyplot as plt

A = np.diag([-1.0, -2.0])
P = np.array([[0.0, 1.0], [1.0, 0.0]])
A0 = A # unswapped field
A1 = P @ A # swapped field
```

```

244 I = np.eye(2)
245
246 def M2(h):
247     """Period-two monodromy for the alternating schedule."""
248     return (I + h*A1) @ (I + h*A0)
249
250 def rho(h):
251     return max(abs(np.linalg.eigvals(M2(h))))
252
253 # --- Reproduce Table 1 -----
254 print(f"{'h':>6} {'trace':>10} {'det':>10} {'rho':>10}")
255 for h in [0.05, 0.20, 0.50, 0.90, 1.00, 1.20]:
256     M = M2(h)
257     print(f"{h:6.2f} {np.trace(M):10.4f} {np.linalg.det(M):10.4f} {
258     rho(h):10.4f}")
259
260 # --- Figure 1: spectral radius vs h -----
261 hs = np.linspace(1e-3, 1.4, 600)
262 rhos = np.array([rho(h) for h in hs])
263 plt.figure()
264 plt.plot(hs, rhos, label=r'$\rho(\mathbf{M}_2)$')
265 plt.axhline(1.0, ls='--', c='k', label=r'$\rho=1$')
266 plt.axvline(1.0, ls=':', c='r', label=r'$h=1$')
267 plt.xlabel('step size h'); plt.ylabel('spectral radius')
268 plt.legend(); plt.tight_layout(); plt.savefig('Fig1.tif', dpi=300)
269
270 # --- Figure 2: trajectory norms at h = 0.5 -----
271 h = 0.5
272 N = 40
273 def run(schedule, x0=np.array([1.0, 1.0])):
274     x = x0.copy(); out = [x.copy()]

```

```

275     for s in schedule:
276         F = A @ x
277         x = x + h * ((1 - s) * F + s * (P @ F))
278         out.append(x.copy())
279     return np.array(out)
280
281 sched_uncoupled = [0]*N
282 sched_full      = [1]*N
283 sched_alternating= [k % 2 for k in range(N)]
284
285 plt.figure()
286 for name, sched in [('uncoupled', sched_uncoupled),
287                     ('fully coupled', sched_full),
288                     ('alternating', sched_alternating)]:
289     traj = run(sched)
290     plt.semilogy(np.linalg.norm(traj, axis=1), label=name)
291 plt.xlabel('step k'); plt.ylabel(r'$\|\mathbf{x}_k\|$')
292 plt.legend(); plt.tight_layout(); plt.savefig('Fig2.tif', dpi=300)
293
294 # --- Figure 3: verification scatter -----
295 plt.figure()
296 plt.scatter(hs, rhos, s=4, label='direct eigenvalue computation')
297 plt.axhline(1.0, ls='--', c='k')
298 plt.xlabel('step size h'); plt.ylabel('spectral radius')
299 plt.legend(); plt.tight_layout(); plt.savefig('Fig3.tif', dpi=300)
300

```

301 Running this script should reproduce the entries of Table 1 to four decimal places. If any entry
302 disagrees, the discrepancy must be resolved before submission.

References

- [1] Butcher, J.C. (2016). *Numerical Methods for Ordinary Differential Equations*, 3rd edn. Wiley, Chichester.
- [2] Gupta, H.K. (1992). *Reservoir-Induced Earthquakes*. Elsevier, Amsterdam.
- [3] Hairer, E., Lubich, C., and Wanner, G. (2006). *Geometric Numerical Integration*, 2nd edn. Springer, Berlin.
- [4] Iserles, A. (2009). *A First Course in the Numerical Analysis of Differential Equations*, 2nd edn. Cambridge University Press, Cambridge.
- [5] Leimkuhler, B., and Reich, S. (2004). *Simulating Hamiltonian Dynamics*. Cambridge University Press, Cambridge.
- [6] Liberzon, D. (2003). *Switching in Systems and Control*. Birkhäuser, Boston.
- [7] McLachlan, R.I., and Quispel, G.R.W. (2002). Splitting methods. *Acta Numerica* **11**: 341–434.
- [8] Scholz, C.H. (1998). Earthquakes and friction laws. *Nature* **391**: 37–42.
- [9] Simpson, D.W. (1976). Seismicity changes associated with reservoir loading. *Engineering Geology* **10**(2–4): 123–150.
- [10] Strang, G. (1968). On the construction and comparison of difference schemes. *SIAM Journal on Numerical Analysis* **5**(3): 506–517.
- [11] Trefethen, L.N., and Bau, D. (1997). *Numerical Linear Algebra*. SIAM, Philadelphia.
- [12] Yoshida, H. (1990). Construction of higher order symplectic integrators. *Physics Letters A* **150**(5–6): 262–268.